

ACCIDENTAL IGNITION OF GAS CYLINDER CONTENTS

Lessons Learned

This Lessons Learned Document focuses on an incident in which the contents of a modified compressed gas cylinder ignited. The small explosion that resulted burned one researcher, caused transient hearing issues in several others, and launched a metal rod 25 feet. It demonstrates the risks of modifying equipment and the importance of proper storage and labeling and of performing a risk assessment (especially before beginning unfamiliar tasks).

What Happened?

After working all afternoon on a laboratory clean-out, lab members found a K size (51 inch tall, or 130 cm) compressed gas cylinder with a label reading:

STANFORD-MADE MIXING TANK

This tank has a steel rod running down the center of it and should be moved with caution. No bumps or excessive jostling.

Maximum pressure in the mixing tank is not to exceed 15 psig

May contain ethylene and/or ethylene/oxygen mixture

Questions? Contact John Doe *[name altered to preserve anonymity]*

While mixtures of flammable and oxidizing gases are not uncommon for the lab given the particular nature of their research, the mention of a metal rod and the uncertainty about the exact contents gave them pause. Two staff scientists assumed responsibility for the cylinder, instructing other researchers not to handle it. They reached out by phone to the listed contact, a staff scientist who had retired. The retired scientist explained that the item was left from a project that had ended roughly 15 years prior. At that time the lab had purchased a used compressed gas cylinder and modified it by adding a metal rod with small holes in it as part of the structure of the cylinder valve stem. The retired scientist could not recall what was in the cylinder and advised that it be decommissioned. Stanford's primary compressed gas providers will not pick up cylinders that they do not own, so the staff scientists agreed in consultation with the retired scientist to turn it into scrap metal by removing the valve stem and metal rod themselves. The retired scientist cautioned that there had been an ignition incident with a similarly modified cylinder years before, an incident which caused the lab to stop using retrofitted gas cylinders as mixing tanks and to instead purchase purpose-built mixing tanks.



The staff scientists first planned to connect the cylinder to a manifold/vacuum pump system to assess the

pressure of the contents and to evacuate any gas contained within. After searching for ~30 minutes, they were unable to find a compatible regulator and decided on an alternative approach. The staff scientists secured the cylinder to a cylinder cart, wheeled it outside, and stationed the cylinder on flat ground. With both staff scientists holding the cylinder, one briefly cracked open the valve and quickly closed it again. Hearing no gas escape, they opened the valve slowly until it was fully open. Again, no gas flow was heard, so the staff scientists closed it. They discussed and assumed that the cylinder had been properly emptied and back-filled with an inert gas per laboratory policy, meaning it was filled with inert gas at ambient pressure.

Operating under this assumption, the staff scientists obtained a wrench and attempted to remove the valve stem but found that they could not complete the task with only two people. They re-capped the cylinder and transported it to a more isolated outdoor location near the lab. They removed the cylinder from the cart and laid it on its side, then removed the cap. The staff scientists enlisted the help of four graduate students. One of the staff scientists and the four graduate students knelt perpendicular to the cylinder, each holding it with two hands. The other staff scientist used the wrench to begin removing the valve. After several turns, the staff scientist was able to pull the valve and attached rod out slightly. They paused, and the lab members who had been holding the cylinder let go and stood up.

The staff scientist continued to pull the rod out of the cylinder and when ~1 foot of the rod was exposed, the contents of the cylinder ignited. There was a loud bang, a bright flame jet emitted from the cylinder, and the metal rod was launched out of the cylinder. All researchers present began checking on each other's status. All reported ringing in their ears. The staff scientist who had been removing the rod reported hearing loss in their left ear and painful scrapes and burns on their right hand. The other staff scientist and one of the graduate students began administering first aid, helping the injured staff scientist to rinse their hand with cool water, apply burn dressing, and wrap the burned area in gauze. Another lab member who had not been present for the incident drove the injured staff scientist to the Stanford Hospital Emergency Department, where they received treatment for first- and second-degree burns to their hands but were not admitted. Meanwhile, the other researchers found the rod, which had flown along the ground ~25 feet (~7.5 meters) away, brought it back to the scene of the incident, and moved the cylinder to a gas storage location. The lab notified EH&S via email and had all researchers who had been involved write down their account of the incident. The uninjured staff scientist used these accounts to write up and submit an incident report. The injured researcher's burns healed, and all researchers' hearing issues resolved without treatment.

Why Did This Happen?

The immediate cause of this incident was most likely sparks from the steel rod scraping against the cylinder wall igniting a flammable gas mixture in the cylinder. While the cylinder contents were no longer under pressure, this did not mean the cylinder was "empty". Based on the cylinder's label, it is reasonable to assume that its contents included ethylene, a flammable gas, in a concentration above its lower explosive limit. For the mixture to ignite, it must also have contained oxygen in some concentration, either from the initial mixture or from the flow of ambient air into the cylinder when the valve was opened and the stem removed.

There were several root causes involved in this incident

- Improper modification of existing equipment for new purposes: Laboratories often face the challenge of reconciling their need for expensive, specialized equipment with a limited budget. This can lead to creative alternative uses for existing items or making modifications to equipment to allow it to serve a new purpose. While this can be acceptable if those involved have the right information, skills, and safety

controls, in other cases it can lead to incidents. For example, equipment purpose-built for working with flammable gases will generally be constructed such that it will not produce sparks, which in this circumstance would have removed the likely source of ignition.

- Inappropriate storage: The lab's research involves the study of reactive gas mixtures, so the mixing of flammable and oxidizing gases is reasonable for experimental purposes. However, such mixtures should not be stored longer than necessary. They should be mixed in the minimum quantity necessary for an experiment and any excess must be vented carefully.
- Lack of formal risk assessment: While the researchers did stop to consider the potential hazards of the cylinder, they did not perform a systematic risk assessment, leading to key risks not being identified. Most notably, had the researchers performed a formal risk assessment, they most likely would have identified the possibility that the cylinder's contents were flammable, and may also have identified the possibility of sparking if the rod contacted the cylinder walls.

Other factors that contributed to the incident include:

- Lack of awareness of EH&S resources: The lab believed that they needed to handle the cylinder's disposal themselves. However, EH&S can assist with the disposal of unique research equipment, containers (including gas cylinders) with unknown contents, and more.
- Inadequate labeling: While the label included possible contents, the lack of certainty left room for interpretation and led the researchers to eventually assume the cylinder no longer contained any flammable gas. Had the label definitively stated that the cylinder contained ethylene, or even definitively stated that the contents were flammable, the researchers would have approached management of the cylinder differently.
- Inadequate self-inspection procedures: Despite the cylinder being present in the lab with an inadequate label for over a decade, no quarterly self-inspections noted this. More thorough self-inspections could have allowed the lab to address the issue much earlier, possibly when there was a better chance a lab member knew the cylinder's contents.
- Incomplete follow-up to previous incidents: The retired researcher noted that there had been an incident in the lab involving accidental ignition of a flammable mixture in a similar "mixing tank." While this led to the lab starting to purchase purpose-built mixing tanks rather than modifying cylinders, existing modified cylinders posing similar risks to the one involved in the incident were not decommissioned.

What Went Right?

Explosions in a research environment can lead to devastating consequences, up to and including death. Some of the steps taken by the researchers prevented more serious consequences and reduced the harm caused by the incident.

- The researchers moved the cylinder to a clean, open location outdoors. If the cylinder had remained indoors, been in a more crowded location, or been in a less clean location, the steel rod could have injured people or destroyed lab property when it launched, or the explosion could have ignited surrounding items.
- The cylinder was laid on its side. This is extremely hazardous to do to a pressurized cylinder (as any damage to the valve can lead to the cylinder being propelled in unpredictable directions), but because the researchers had verified it was not pressurized, this was an acceptable course of action. Importantly, this action meant that the explosion launched the rod in a direction free of any people. Had the cylinder

been upright, one or more researchers could have had body parts in the flight path, or been standing where the rod was likely to land, which could have resulted in serious injury or death.

- Lab members present at the incident responded excellently. The injured researcher received appropriate first aid, and a lab member not involved in the incident (who thus had no hearing issues related to the incident and was in a calmer mental state) transported the injured researcher. Other lab members documented the scene of the incident. All involved wrote down accounts of the incident as soon as possible and used this information to submit a detailed incident report.
- The lab performed a good internal investigation, seeking to identify not just the immediate causes of the incident but the root causes as well. As a result of their findings the lab identified and implemented a number of procedural improvements on their own, in addition to other measures recommended by EH&S.

How to Prevent and Respond to Accidents Like This

EH&S has a number of recommendations and resources for preventing both incidents like this and incidents in general. The lab also devised and implemented a number of improvements in their practices and operating procedures.

- Perform and document a risk assessment before performing a new experiment, making significant alterations to an existing procedure, or performing other novel tasks in the laboratory. Completing and documenting a formal risk assessment helps researchers to take a methodical, systematic view of their proposed plans, offering an opportunity to catch invalid assumptions and identify ways to protect against hazards. EH&S recommends using the Laboratory Risk Assessment Tool and is always available for consultation regarding both when a formal risk assessment is appropriate and the contents of such an assessment.
- Purchase purpose-built equipment from reputable vendors, rather than modifying existing equipment or constructing something oneself. This should be especially prioritized for equipment that will be used with hazardous materials and/or conditions; in some circumstances, use of certified equipment may be mandated by law. For assistance in determining when “home built” equipment would be safe and appropriate, consult EH&S.
 - Altering existing equipment to use for unintended purposes can void the warranty. Consult the manufacturer before altering equipment.
 - The lab involved in this incident formalized their policy that mixing tanks must be purpose-built and assessed for their operating ranges and failure modes.
- Conduct thorough self-inspections on (at minimum) a quarterly basis. If you observe issues, either correct them or consult with the person responsible to ensure they correct the issue. Pay special attention to areas that are used infrequently.
 - Correcting issues such as poor labeling should not be limited to self-inspections. Lab members should always be alert for issues and if they find any should correct them, contact the lab safety coordinator, or contact EH&S.
- Always label containers with the full chemical name of the contents, as well as the concentration(s) and hazard information.
 - The laboratory involved in this incident instituted a rule that all mixing tanks must be labeled with

the exact contents of the tank (including common name, chemical formula, and concentration for all constituents), the approximate pressure of the tank, the name and contact information of the person responsible for the mixture, the date the mixture was created, and the anticipated dates the mixture will be used.

- Establish clear expectations for acceptable states for equipment. Always return equipment to its “at rest” state after use.
 - The lab involved in this incident instituted a lab rule that mixing tanks are only permitted to be in two states. The first acceptable state is active use, which means that the tank is actively being used in an experiment. Such tanks must be labeled as noted above. The second state is inactive, meaning the tank has either been flushed with an inert gas or evacuated and left under vacuum or refilled with inert gas.
- Perform thorough follow-up after incidents. A lab’s principal investigator (PI) should conduct a root cause analysis after any incident and should identify and lead implementation of corrective actions. EH&S is available to assist with these tasks.
 - A similar incident had occurred previously. At that time, the lab discontinued use of similar mixing tanks (such as the one involved in this more recent incident) but did not dispose of all the tanks they had at that time.
 - After this incident, the lab reviewed all gas cylinders and mixing tanks in the lab and worked with EH&S to dispose of items of concern.
- Implement clear expectations and procedures for researchers leaving the lab. Departing researchers should ensure that all unneeded materials are disposed of and that all remaining materials are clearly and correctly labeled.
- When in doubt, such as when approaching unclearly labeled items or handling unusual equipment, contact EH&S before proceeding. EH&S employs a wide range of experts who can assist you in completing a wide variety of tasks safely, including disposal of unusual items and assessing the risk of unfamiliar situations.
- Conduct lab clean-outs, particularly hazardous operations, and other tasks where EH&S assistance is more likely to be needed during normal business hours to ensure prompt response.
- Report serious incidents and incidents requiring EH&S assistance by calling 650-725-9999. This number is monitored at all times and is the best way to receive prompt EH&S assistance. Incidents involving hospitalization must be reported as soon as possible, no later than 8 hours after the incident. All incidents must also be reported via an incident report form.